

Maven Project To Connect To Web Servers Using Selenium Dependency

Maven Project with Multiple Websites Automation Under Firefox Browser

This Maven project uses Selenium and Apache Maven to automate testing of multiple websites using the Firefox browser.

In this project, Selenium WebDriver with GeckoDriver is used to launch Mozilla Firefox and connect to multiple websites automatically. Maven manages all Selenium dependencies and project configurations through the pom.xml file.

Websites Included in the Project

1. [Sauce demo website](#)
2. [Automation Exercise](#)
3. [Practice Test Automation Website](#)

Working Process

1. Maven automatically downloads Selenium libraries.
2. GeckoDriver launches the Firefox browser.
3. Selenium WebDriver connects to the websites one by one.
4. Automated actions such as login testing, navigation, form validation, and search operations are performed.
5. Results are verified and displayed automatically.

Features

- Multiple website automation in one project
- Firefox browser automation using GeckoDriver
- Automated web testing
- Maven dependency management
- CI/CD support for DevOps environments

Applications

- Functional testing
- Regression testing
- Browser automation
- Continuous testing in DevOps pipelines

Advantages

- Reduces manual testing effort
- Faster and reliable testing
- Easy integration with automation tools
- Improves software quality
- Supports tools like Jenkins for continuous integration

This project demonstrates how Selenium and Maven can automate multiple websites efficiently using the Firefox

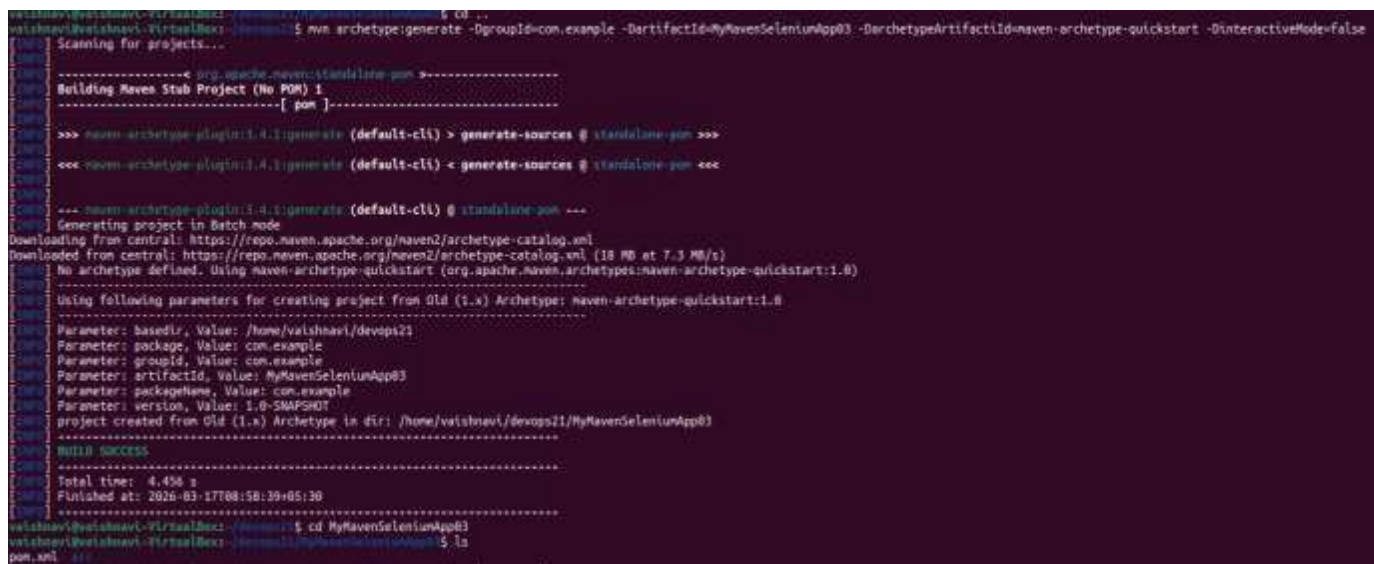
browser in software testing and DevOps workflows.

Steps to execute the project:

Step1:

To create a basic Maven Java project, run the following command:

```
$mvn archetype:generate -DgroupId=com.example -
-DartifactId=MyMavenSeleniumApp05 -DarchetypeArtifactId=maven-archetype-
quickstart -DinteractiveMode=false
```



```
vaishnav@vaishnavi-VirtualBox: /devops21 $ cd ..
vaishnav@vaishnavi-VirtualBox: /devops21 $ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp03 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] [ pom ]-----
[INFO]
[INFO] >>> mvn-archetype-plugin:1.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO] <<< mvn-archetype-plugin:1.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- mvn-archetype-plugin:1.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 7.3 MB/s)
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] -----
[INFO] Parameter: basedir, Value: /home/vaishnavi/devops21
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp03
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/vaishnavi/devops21/MyMavenSeleniumApp03
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 4.456 s
[INFO] Finished at: 2025-03-17T08:58:05:30
[INFO]
[INFO] -----
vaishnav@vaishnavi-VirtualBox: /devops21 $ cd MyMavenSeleniumApp03
vaishnav@vaishnavi-VirtualBox: /devops21/MyMavenSeleniumApp03 $ ls
pom.xml
```

The command can be broken down as:

- DgroupId=com.example – project’s base package (reverse-domain style).
- DartifactId=my-maven-app – defines the name of the project (will also be the folder name).
- DarchetypeArtifactId=maven-archetype-quickstart – sets the Maven template to be the basic Java application.
- DinteractiveMode=false – deactivates interactive mode and uses the given values automatically.

Step 2:

Use \$cd command to go into the directory.

Execute \$tree command to view the structure of the project

Step 3:

Edit the pom.xml file

Add the required dependencies and plugins required from central maven

respository For given project, we will download unit testing dependency from

Jenkins.

```
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.example</groupId>
<artifactId>MyMavenSeleniumApp05</artifactId>
<packaging>jar</packaging>
<version>1.0-SNAPSHOT</version>
<name>MyMavenSeleniumApp05</name>
<url>http://maven.apache.org</url>
<dependencies>
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>3.8.1</version>
<scope>test</scope>
</dependency>
<!-- https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
<dependency>
<groupId>org.seleniumhq.selenium</groupId>
<artifactId>selenium-java</artifactId>
<version>4.29.0</version>
</dependency>
</dependencies>
<!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>11</source>
```

```
<target>11</target>

</configuration>

</plugin>

<plugin>

  <groupId>org.apache.maven.plugins</groupId>

  <artifactId>maven-shade-plugin</artifactId>

  <version>3.2.4</version>

  <executions>

    <execution>

      <phase>package</phase>

      <goals>

        <goal>shade</goal>

      </goals>

      <configuration>

        <transformers>

          <transformer
implementation="org.apache.maven.plugins.shade.resource.ManifestResourceTransformer">

            <mainClass>com.example.App</mainClass>

          </transformer>

        </transformers>

      </configuration>

    </execution>

  </executions>

</plugin>

<!-- Example: Maven Surefire Plugin to run tests -->

<plugin>

<groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-surefire-plugin</artifactId>

<version>2.22.2</version>

</plugin>

<plugin>

  <groupId>org.apache.maven.plugins</groupId>

  <artifactId>maven-jar-plugin</artifactId>
```

```

<version>3.0.1</version>

<configuration>

  <archive>

    <manifestEntries>

      <Main-Class>com.example.App</Main-Class>

    </manifestEntries>

  </archive>

</configuration>

</plugin>

</plugins>

</build>

</project>

```

Step 4:

Execute the command \$mvn compile

If the command shows output as “build success”, it means that the project has compiled successfully

```

[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyHavenApp >-----
[INFO] Building MyHavenApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyHavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vaishnavi/devops21/MyHavenApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyHavenApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.713 s
[INFO] Finished at: 2026-03-24T21:20:16+05:30
[INFO]

```

Step 5:

- Right-click on the webpage and select Inspect.
- The HTML code will be displayed. Click on the cursor icon (arrow symbol) available on the top-left of the inspect panel, then hover over the username field to identify its ID, which is user-name.
- Similarly, check the search button (ID:submit_search). Make a note of these IDs, as they will be used in the App.java file.
- Next, Edit the

App.java file package

com.example;

import org.openqa.selenium.By;

```

import org.openqa.selenium.WebDriver;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.FirefoxOptions;

public class App {
    public static void main(String[] args) {

        // 🟡 Setup Firefox for Jenkins (Headless mode)
        FirefoxOptions options = new FirefoxOptions();
        options.addArguments("--headless");           // REQUIRED
        options.addArguments("--no-sandbox");         // Recommended
        options.addArguments("--disable-dev-shm-usage"); // Recommended

        WebDriver driver = new

        FirefoxDriver(options); try {
            // ---- Test 1: SauceDemo Login ----
            driver.get("https://www.saucedemo.com/");
            driver.findElement(By.id("user-
            name")).sendKeys("standard_user");
            driver.findElement(By.id("password")).sendKeys("secret_sauce"
            ); driver.findElement(By.id("login-button")).click();
            // ---- Test 2: Practice Test Login ----
            driver.get("https://practicetestautomation.com/practice-test-login/");
            driver.findElement(By.id("username")).sendKeys("student");
            driver.findElement(By.id("password")).sendKeys("Password123");
            driver.findElement(By.id("submit")).click();

            // ---- Test 3: Product Search ----
            driver.get("https://automationexercise.com/products");
            driver.findElement(By.id("search_product")).sendKeys("blue top");
            driver.findElement(By.id("submit_search")).click();

            System.out.println("All test cases executed successfully 🟢 ");

        } catch (Exception e) {
            System.out.println("Error occurred: " + e.getMessage());
            e.printStackTrace();
        } finally {
            // Close browser
            driver.quit();
        }
    }
}

```

Step 6:

Build and run the maven application Execute the command \$mvn clean install

```

valshen@valshen-virtualbox: ~/devops21/PracticeTest2
$ mvn clean compile
[INFO] Scanning for projects...
[INFO]
[INFO] ----- [com.example.seleniumapp:MyMavenSeleniumApp 1.0-SNAPSHOT] -----
[INFO]
[INFO] --- maven-clean-plugin:3.1.0:clean (default-clean) @ MyMavenSeleniumApp ---
[INFO] Deleting /home/valshen/devops21/PracticeTest2/target
[INFO]
[INFO] --- maven-resources-plugin:3.1.0:resources (default-resources) @ MyMavenSeleniumApp ---
[INFO] Using platform encoding (UTF-8 actual) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/valshen/devops21/PracticeTest2/src/main/resources
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenSeleniumApp ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 1 source file to /home/valshen/devops21/PracticeTest2/target/classes
[INFO]
[INFO] --- maven-resources-plugin:3.1.0:resources (default-resources) @ MyMavenSeleniumApp ---
[INFO] Using platform encoding (UTF-8 actual) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/valshen/devops21/PracticeTest2/src/test/resources
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenSeleniumApp ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 1 source file to /home/valshen/devops21/PracticeTest2/target/classes
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenSeleniumApp ---
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.example.AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenSeleniumApp ---
[INFO] Building jar: /home/valshen/devops21/PracticeTest2/target/MyMavenSeleniumApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:3.1.0:install (default-install) @ MyMavenSeleniumApp ---
[INFO] Installing /home/valshen/devops21/PracticeTest2/target/MyMavenSeleniumApp-1.0-SNAPSHOT.jar to /home/valshen/.m2/repository/com/example/MyMavenSeleniumApp/1.0-SNAPSHOT/MyMavenSeleniumApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:3.1.0:install (default-install) @ MyMavenSeleniumApp ---
[INFO] Installing /home/valshen/devops21/PracticeTest2/target/MyMavenSeleniumApp-1.0-SNAPSHOT.jar to /home/valshen/.m2/repository/com/example/MyMavenSeleniumApp/1.0-SNAPSHOT/MyMavenSeleniumApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.001 s
[INFO] Finished at: 2020-10-20T11:11:40-0400
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenSeleniumApp ---

```

Step 7:

Execute `$ java -jar target/MyMavenSeleniumApp03-1.0-SNAPSHOT`

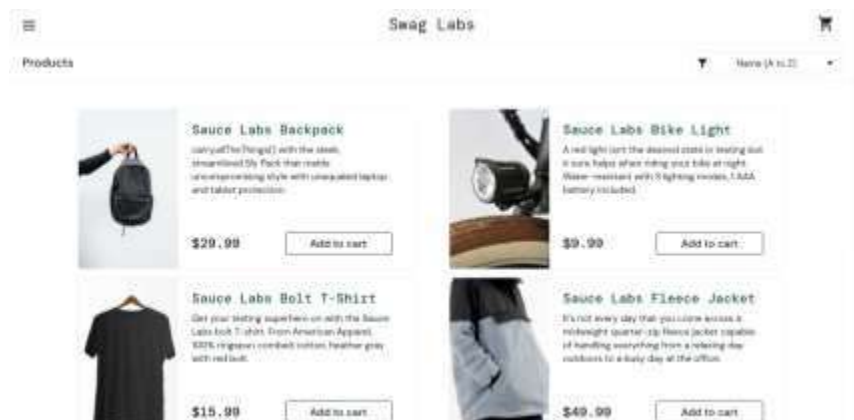
If we get error in executing the above jar file do run the below command

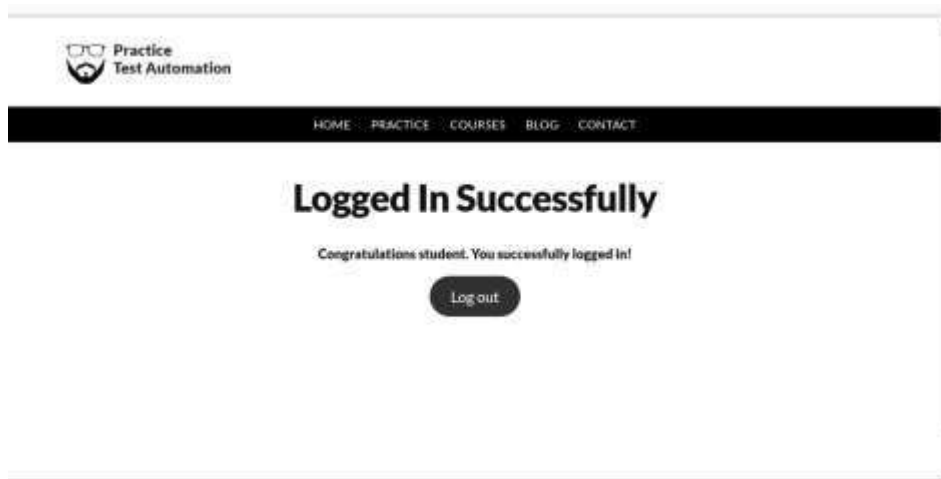
`$mvn exec:java -Dexec.mainClass="com.example.App"`

```

valshen@valshen-virtualbox: ~/devops21/PracticeTest2
$ mvn exec:java
[INFO] Scanning for projects...
[INFO]
[INFO] ----- [com.example.seleniumapp:MyMavenSeleniumApp 1.0-SNAPSHOT] -----
[INFO]
[INFO] --- maven-clean-plugin:3.1.0:clean (default-clean) @ MyMavenSeleniumApp ---
[INFO] Deleting /home/valshen/devops21/PracticeTest2/target
[INFO]
[INFO] --- maven-resources-plugin:3.1.0:resources (default-resources) @ MyMavenSeleniumApp ---
[INFO] Using platform encoding (UTF-8 actual) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/valshen/devops21/PracticeTest2/src/main/resources
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenSeleniumApp ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 1 source file to /home/valshen/devops21/PracticeTest2/target/classes
[INFO]
[INFO] --- maven-resources-plugin:3.1.0:resources (default-resources) @ MyMavenSeleniumApp ---
[INFO] Using platform encoding (UTF-8 actual) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/valshen/devops21/PracticeTest2/src/test/resources
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenSeleniumApp ---
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.example.AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenSeleniumApp ---
[INFO] Building jar: /home/valshen/devops21/PracticeTest2/target/MyMavenSeleniumApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:3.1.0:install (default-install) @ MyMavenSeleniumApp ---
[INFO] Installing /home/valshen/devops21/PracticeTest2/target/MyMavenSeleniumApp-1.0-SNAPSHOT.jar to /home/valshen/.m2/repository/com/example/MyMavenSeleniumApp/1.0-SNAPSHOT/MyMavenSeleniumApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:3.1.0:install (default-install) @ MyMavenSeleniumApp ---
[INFO] Installing /home/valshen/devops21/PracticeTest2/target/MyMavenSeleniumApp-1.0-SNAPSHOT.jar to /home/valshen/.m2/repository/com/example/MyMavenSeleniumApp/1.0-SNAPSHOT/MyMavenSeleniumApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.001 s
[INFO] Finished at: 2020-10-20T11:11:40-0400
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenSeleniumApp ---

```





Pushing The Project into git

Execute the commands:

```
$ git config --global user.name "vaishnavim1121"
```

```
$ git config --global user.email "vaishnavim1121@gmail.com"
```

```
$ git init
```

```
$ git add .
```

```
$ git commit -m "initial commit"
```

```
$ git remote add origin https://github.com/vaishnavim1121/MyMavenSeleniumFirefox.git
```



```
$ git push -u origin master
```

Second Phase

Follow the given steps to deploy the project on Jenkins.

go to the Jenkins home page and click on “New Item” to create a job, enter the item name select Pipeline as the project type.

This option allows you to build, test, and deploy applications using a Jenkinsfile. After selecting the pipeline type, click “OK” to proceed to the job configuration page

Definition

Pipeline script from SCM

SCM

Git

Repositories

Repository URL

https://github.com/vaishnavim1121/MyMavenSeleniumFirefox

Please enter Git repository.

Credentials

github credentials

+ Add

Advanced

Save Apply

Now we need to add a Jenkinsfile into our project which will consist of the operations that will occur when building the project in Jenkins

It includes step wise instructions guiding at each stage with right commands

```

pipeline {
    agent any // use any available agent

    tools {
        maven 'Maven' // Ensure this matches the name configured in Jenkins
    }

    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url: 'https://github.com/vaishnavim1121/MyMavenSeleniumApp01.git'
            }
        }

        stage('Build') {
            steps {
                sh 'mvn clean package' // Run Maven build
            }
        }

        stage('Test') {
            steps {
                sh 'mvn test' // Run unit tests
            }
        }

        stage('Run Application') {
            steps {
                // Start the JAR application
                sh 'java -jar target/MyMavenSeleniumApp01-1.0-SNAPSHOT.jar'
            }
        }
    }
}

```

Now try building the project with Jenkins

Activities Google Chrome May 9 19:24

valshnavim1121/Gradi... MyMavenSeleniumFirefox

localhost:8080/job/MyMavenSeleniumFirefox/

Jenkins MyMavenSeleniumFirefox

Status MyMavenSeleniumFirefox Add description

Changes

Build Now

Delete Pipeline

Configure

Rename

Full Stage View

Stages

Pipeline Syntax

Built >

Filter

13 April 2026

#3 22:02

#2 21:58

#1 21:58

Average stage times: (Full run time: ~28s)

	Declarative: Checkout SCM	Declarative: Tool Install	Checkout	Build	Test	Run Application	Declarative: Post Actions
Apr 12 22:02	1s	85ms	922ms	7s	2s	16s	77ms
Apr 12 21:58	1s	98ms	982ms	8s	3s	22s	88ms
Apr 12 21:58	2s	89ms	982ms	9s	2s	482ms	88ms

Permalinks

- Last build (#3), 26 days ago
- Last stable build (#3), 26 days ago
- Last successful build (#3), 26 days ago
- Last failed build (#2), 26 days ago

The error was due to failure in github repository, by directing the correct link to the Jenkins, and re-building it, the project is built successfully,